ETHANOL AND ETHENE

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Ethanol (ethyl alcohol, CH_3CH_2OH) is a clear, colourless liquid. It is miscible (mixes) with water and burns forming carbon dioxide and water.

1 Write a balanced formula equation for the combustion of ethanol.

Ethanol can be added to petrol (up to 10%) to form **gasohol**. Ethanol is a very important solvent and a raw material for other industrial chemicals.

Ethene (ethylene, CH_2CH_2) is a flammable, colourless gas that is insoluble in water.

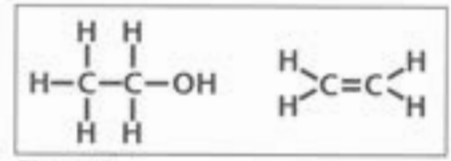


Figure 1

2 Figure 1 shows the structural formulae of ethanol and ethene. Copy figure 1, name each molecule and give its molecular formula.

Ethene is used to make polythene, ethanol and 1, 2-ethanediol (ethylene glycol, 'antifreeze'). Figure 2 shows the structural formula of 1, 2-ethanediol.

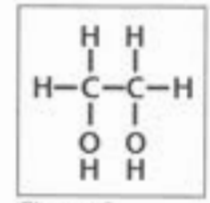


Figure 2

3 Copy figure 2. Name the molecule and give the molecular formula.

Ethanol is made industrially by two processes: anaerobic (no oxygen present) fermentation of carbohydrates and by the hydration (addition of water) of ethene. All ethanol used in beverages (drinks) is obtained by fermentation. Raw materials used for fermentation include grains, potato and sugar. Fermentation is an enzyme-controlled reaction. In the presence of yeast, glucose ($C_6H_{12}O_6$) is fermented to ethanol and carbon dioxide gas. The yeast contains enzymes (organic catalysts) that enable the reaction to take place at room temperature. The maximum ethanol concentration obtained by fermentation is around 15% – any higher than this and the yeast is poisoned by the ethanol. Ethanol can be obtained from an ethanol/water mixture by fractional distillation.

4 Write a balanced formula equation for the fermentation of glucose solution to ethanol and carbon dioxide.

Most industrial ethanol is made from ethene because it is cheaper to make it this way. The process involves the acid-catalysed hydration of ethene. The catalyst is phosphoric acid (H_3PO_4).

5 Write a balanced formula equation for the hydration of ethene to ethanol.

Ethene is produced by the catalytic cracking (breaking up) of hydrocarbons obtained from petroleum (crude oil). Figure 3 shows how ethene and hydrogen can be produced by the cracking of ethane.

6 Copy figure 3 and write a balanced formula equation for this reaction.

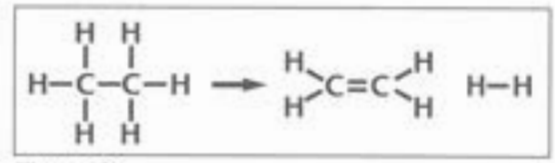


Figure 3

7 Give three uses of ethanol.

8 Give three uses of ethene.

9 Name the raw materials used in the formation of ethanol by fermentation.

10 Explain what is meant by the each term.

- a Miscible
- b Gasohol
- c Anaerobic
- d Hydration
- e Enzyme
- f Cracking
- g Fermentation

11 Give the formula for phosphoric acid.

12 Why is there a limit to the percentage of ethanol in a solution formed by fermentation?

13 Name the process used to extract ethanol from an aqueous solution.

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